RETAILMART V3 · SQL

JOINs Part 1 — Quick Quiz

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STORY

Answer Before Looking!

For each question, try to answer BEFORE scrolling to the answer. This builds interview confidence.

Quiz 1: INNER vs LEFT — What Changes?

EASY

Query A: SELECT c.*, COUNT(o.order_id) FROM customers c INNER JOIN orders o ON c.customer_id = o.cust_id GROUP BY c.customer_id

Query B: Same but with LEFT JOIN instead of INNER JOIN.

ANSWER: Query A shows only customers who have placed at least 1 order (INNER drops unmatched). Query B shows ALL customers — those with zero orders have COUNT = 0. LEFT JOIN never drops left-side rows.

Quiz 2: Why COUNT(*) vs COUNT(o.order_id)?

MEDIUM

In a LEFT JOIN customers to orders, what does COUNT(*) return for a customer with NO orders? What about COUNT(o.order_id)?

ANSWER: COUNT(*) = 1 (the customer row itself exists). COUNT(o.order_id) = 0 (order_id is NULL for unmatched rows, and COUNT skips NULLs). ALWAYS use COUNT(right_table.col) in LEFT JOINs.

Quiz 3: What Happens to the LEFT JOIN?

MEDIUM

ANSWER: The LEFT JOIN silently becomes an INNER JOIN! WHERE o.net_total > 10000 removes all rows where net_total IS NULL — which includes ALL customers with no orders. Fix: put the filter in ON: LEFT JOIN … ON … AND o.net_total > 10000.

Question

✅ ANSWER

```
SELECT c.customer_id, o.net_total
FROM customers.customers c
LEFT JOIN sales.orders o
  ON c.customer_id = o.cust_id
WHERE o.net_total > 10000;
```

Quiz 4: Cartesian Product Trap

HARD

ANSWER: This is a Cartesian product — every order paired with every customer! 150,000 × 50,000 = 7.5 BILLION rows. This will crash your database. The comma syntax is an implicit CROSS JOIN. ALWAYS use explicit JOIN with ON.

Question

✅ ANSWER

```sql
-- WARNING: read, do NOT run this one!
SELECT * FROM sales.orders, customers.customers;
```

Quiz 5: How Many Rows?

CRAZY

Table A has 100 rows. Table B has 50 rows. 30 rows match on the join key. Some A rows match multiple B rows (total 45 matches).

INNER JOIN rows? 45 (only matches, including duplicates from one-to-many).
LEFT JOIN rows? 45 + 70 = 115 (45 matched + 70 unmatched A rows with NULLs).
RIGHT JOIN rows? 45 + 20 = 65 (45 matched + 20 unmatched B rows with NULLs).